Demystifying the potential pharmaceutical attributes of corn silk: An agriculture bio-waste

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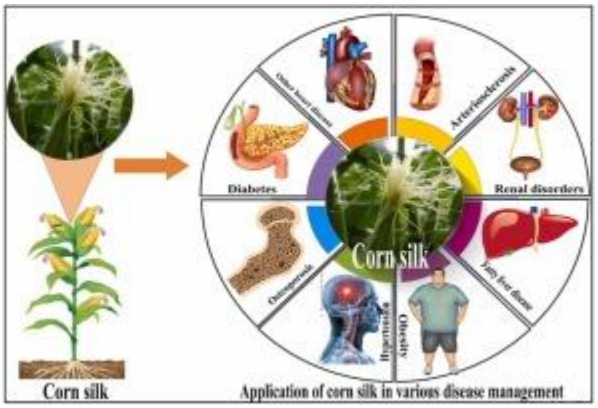
Abstract

Corn silk (CS) typically considered an agricultural by-product, has emerged as a versatile biomimetic resource with considerable potential in the pharmaceutical and therapeutic fields. CS is a plethora of bioactive reservoirs, including alkaloids, flavonoids, polysaccharides, phenolic compounds, and essential vitamins, hence showing a wide range of pharmacological properties. The established antioxidant, anti-inflammatory, antimicrobial, and antidiabetic activities of the compound reflect the possible uses in controlling chronic diseases. Apart from its therapeutic application, CS is an excellent pharmaceutical excipient with remarkable utility. Several literature reports showed that it has been used as a binder, disintegrant, and stabilizer in pharmaceutical formulations. Along with its biocompatibility and eco-friendliness, it makes it a sustainable alternative to synthetic excipients. The valorization of CSs and isolation of bioactive compounds extends its versatility to advanced drug delivery systems, including nanoparticles, liposomes, and controlled-release formulations, where it improves drug solubility, absorption, and bioavailability. Despite its promise, challenges exist, such as variability in chemical composition, formulation stability, large-scale processing, and regulatory hurdles, which makes its full-scale adoption to the mainstream a challenge. With the development of nanotechnology, analytical methods, and personalized medicine in the future, the CS may become a cornerstone of pharmaceutical innovation and sustainable development. CS presents a promising natural resource with multifunctional pharmaceutical applications, warranting further exploration and integration into modern drug delivery and therapeutic systems.

Graphical Abstract



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Introduction

The multidisciplinary field of biomimetics, which draws inspiration from nature's blueprint for developing novel solutions, has revolutionized the pharmaceutical and therapeutic industries (Ciulla et al., 2023). Scientists and researchers work to develop materials, methods, and procedures that can improve medication delivery, improve treatment results, and minimize or even completely eradicate adverse effects by taking inspiration from the dynamics of natural systems. Advanced drug delivery technologies, tissue engineering, and innovative excipient discoveries that connect with the biological system of the human body have all been made possible by biomimetic concepts (Pavan Kalyan & Kumar, 2022). This strategy has great potential to solve the shortcomings of the pharmaceutical formulation, particularly its poor solubility, limited bioavailability, and non-specific drug targeting.

Biomimetics' dependence on natural products as inspiration is one of its core tenets. Natural ingredients have been an essential part of pharmaceutical formulation and medication development for ages. These chemicals serve as excipients, which are inert molecules that aid in medication transport, stability, and effectiveness, in addition some of them exhibit to medicinal property too. Additionally, because they are environmentally benign, biocompatible, and biodegradable, polysaccharides, proteins, and lipids are favored (Santos et al., 2019). The fact that phytochemicals can occasionally provide extra therapeutic or protective benefits, enhancing the formulation, is another justification for considering them as functional excipients.

CS, a by-product of maize farming, is one of the numerous natural resources that has garnered interest due to its possible use in medicine (Singh et al., 2022). Because of its diuretic, anti-inflammatory, qualities, maize silk, the long, thread-like stigmas found at the tops of maize ears, has been utilized in traditional medicine. According to recent studies, maize silk is a great source of bioactive phytochemicals that support its biological activities, including flavonoids, alkaloids, polysaccharides, tannins, and terpenoids (Singh et al., 2022). These characteristics make maize silk a viable option for developing novel excipients that serve two purposes: facilitating medication distribution and supplying pharmacological action. Its polysaccharide structure, excellent film-forming, gelation, and stabilizing qualities make it potentially suitable for use in drug delivery systems, such as hydrogels and nanoparticles. In the context of topical uses, including wound healing, anti-inflammatory treatments, and usage in cosmetic formulations, its phytochemical makeup and antibacterial elements may offer further usefulness. Therefore, the characteristics of maize silk can be connected to the existing trends for multifunctional natural excipients with pharmaceutical systems.

One of the most extensively grown cereal crops in the world, corn is used to produce food, feed, and biofuel (Details for: Conversion of Corn Waste into Value Added Products / › Central Library, University of Rajshahi Catalog, n.d.). At the same time, a significant amount of agro-waste, such as husks, cobs, stalks, and silk, is produced during the harvesting of corn (Adetunji et al., 2022). These include CS, which are fine, thread-like fibers that have drawn a lot of attention lately due to their bioactive properties and potential uses in cosmetics, pharmaceuticals, and functional foods. The materials were either discarded or fed to animals for many years. Recently, this has emerged as a significant material resource for circular economy adoption and sustainable development.

Bioactive substances, including flavonoids, alkaloids, polysaccharides, vitamins, and vital minerals, are found in CS (Mohanty et al., 2022). These substances have strong antibacterial, diuretic, anti-inflammatory qualities. CS hence has a lot of potential as a component in cosmetics, pharmaceuticals, and nutraceuticals. Since maize silk is being converted into a valuable commodity, its valuation is in line with waste reduction, resource efficiency, and environmental sustainability. Thus, the present review elaborates briefly varied potential pharmaceutical attributes of CS.

Section snippets

Composition and bioactive constituents of corn silk

One of the most significant byproducts of maize agriculture is CS which is made up of the silky thread-like stigmas of maize. Despite its long history of usage in traditional medicine, recent research has uncovered its chemical makeup and bioactive components (Yadav et al., 2021). For pharmacological and therapeutic purposes, this makes it a valuable resource. Its chemical makeup, the therapeutic value of its phenolic compounds, techniques for separating its bioactive ingredients, and a

Therapeutic attributes of corn silk

Corn silk is known from ancient traditional medicine yet has gained recent scientific interest as a promising plant for therapeutic properties (Bhatia et al., 2024). The anti-inflammatory, antimicrobial, antidiabetic, and other health-promoting effects of CS will be discussed here, pointing to mechanisms of action and applications in chronic disease management and therapy Fig. 1.

Pharmaceutical applications of corn silk

Corn silk has demonstrated significant potential as a versatile material for use in medical applications. It is a sustainable substitute for standard excipients and innovative formulations because of its particular phytochemical makeup, biocompatibility, and functional properties (Sawangwong et al., 2024a). This section discusses its activities as a pharmaceutical excipient, its usefulness in advanced drug delivery systems, its ability to enhance bioavailability, and its use in topical and

Biomimetic approaches using corn silk

Pharmaceutical research and development are increasingly finding that biomimetics, the discipline of drawing inspiration from natural systems, processes, and structures to address human problems, is an essential field. These sophisticated solutions are suitable models for the most effective and sustainable technology that humans have ever encountered since they have been refined over millions of years in nature. Biomimetics has made it possible for the pharmaceutical industry to develop

Economic and environmental benefits of using corn silk

Including CS in a circular economy model encourages environmentally friendly production methods, lessens reliance on non-renewable resources, and supports sustainable agriculture. Utilizing maize silk lowers environmental pollution and agro-waste disposal expenses. By incorporating CS valorization into their supply chains, agro-industries can reduce resource waste (Mihali et al., 2024). Products made from CS give farmers and agro-industries new sources of income. The markets for bioactive

Challenges and limitations in using corn silk

Long acknowledged for its potential in traditional medicine and as a source of bioactive substances, maize silk is the long, thread-like filaments that grow at the top of an ear of corn. Their pharmacological characteristics, such as their diuretic, anti-inflammatory, and antioxidant effects, have recently attracted attention. Despite maize silk's excellent potential for usage in industrial and pharmaceutical applications, there are still a lot of obstacles and restrictions in its use. The

Future prospects and research directions

The fibrous threads known as CS, which emerge from the top of corn ears, have garnered a lot of interest recently because of their possible medical uses. CS has long been utilized in traditional medicine and has demonstrated potential as a natural treatment for a variety of illnesses, including hypertension, urinary tract infections, and inflammation. There is interest in increasing its uses in medications, functional foods, and industrial items as a result of extensive research into its

Conclusion

Corn silk is a resource that, traditionally considered as an agricultural byproduct, has proven to be a powerful biomimetic resource with an extensive range of applications in pharmaceuticals and therapeutics. The diverse phytochemical profile of CS, encompassing alkaloids, flavonoids, polysaccharides, phenolic compounds, and essential vitamins, forms the basis for its therapeutic potential. It shows strong antioxidant, anti-inflammatory, antimicrobial, antidiabetic, and other health-promoting

CRediT authorship contribution statement

Akshay Kundale: Visualization, Validation, Formal analysis. **Mali Sachin S.:** Writing – original draft, Supervision, Conceptualization. **Dipak S Thorat:** Writing – original draft, Methodology, Data curation. **Sudarshan Singh:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Yogesh V Ushir:** Visualization, Formal analysis. **Prajakta R Patil:** Writing – original draft, Data curation. **Dhanashree R Davare:** Writing – original draft, Data curation. **Pooja V Nagime:**

Author contribution

Sachin S Mali and Sudarshan Singh: Conceived and designed article; Dipak S Thorat: Analyzed and wrote the paper. Yogesh V. Ushir, Prajakta R Patil, and Dhanashree R. Davare: Analyzed, interpreted the data, and wrote the paper. Pooja V Nagime: Analyzed and formal analysis.

Ethical approval

Not applicable

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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